

CCPP Piping Construction Guidebook

*Turkistan Combined Cycle
Power Plant*

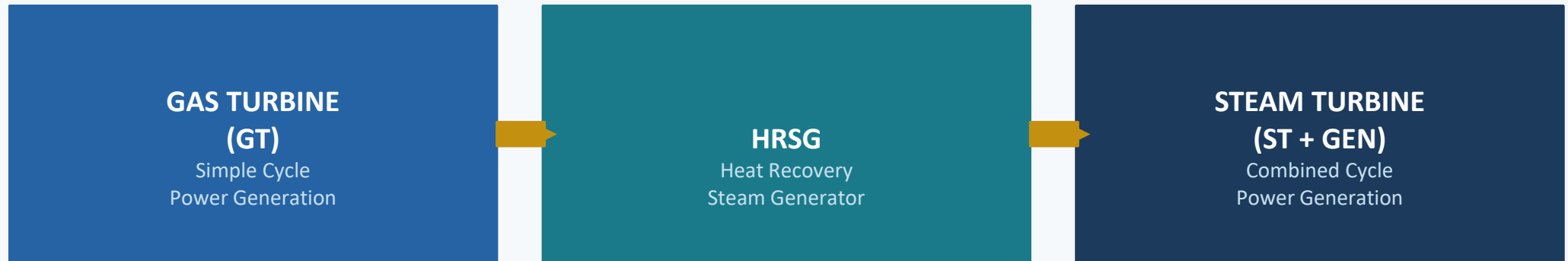
Piping Construction Department
For Stakeholder & New Joiner Orientation

Presentation Scope

- 01 CCPP System Overview & Plant Structure
- 02 Project Timeline & Key Milestone Schedule
- 03 Commissioning Phase Timeline (Detail)
- 04 Piping Completion Priority by Milestone
- 05 Utility Systems (incl. Raw & Potable Water)
- 06 Systems Required Before GT Firing
- 07 Aux. Boiler & Hot Water Supply Concept
- 08 Construction Management & Piping Methods
- 09 Pressure Testing & PWHT Requirements
- 10 Lessons & Learned

Combined Cycle Power Plant (CCPP) — System Overview

Integrated power generation: GT exhaust heat drives steam turbine for additional power generation



>55%

Overall
Efficiency

HP·LP

Steam Pressure
Levels

100%

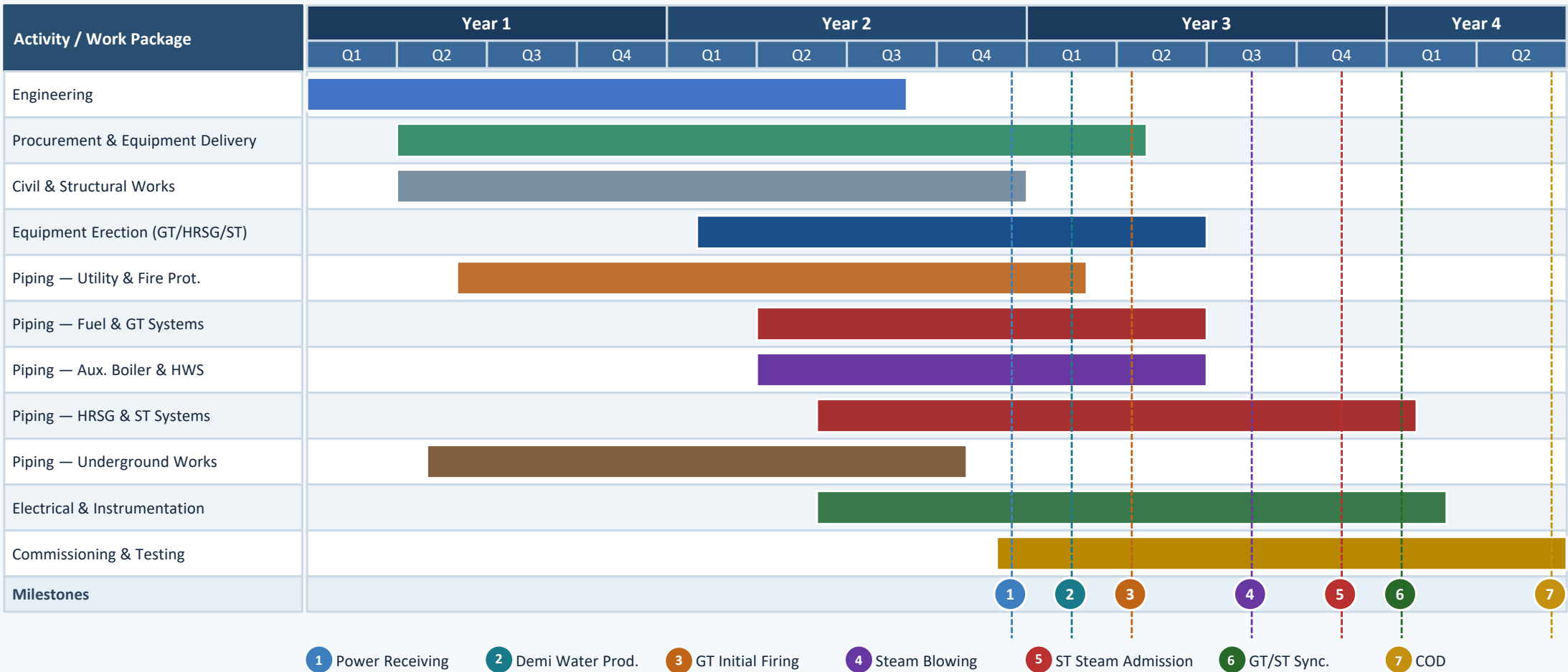
NDE on
Critical Welds

ASME B31.1

Governing
Piping Code

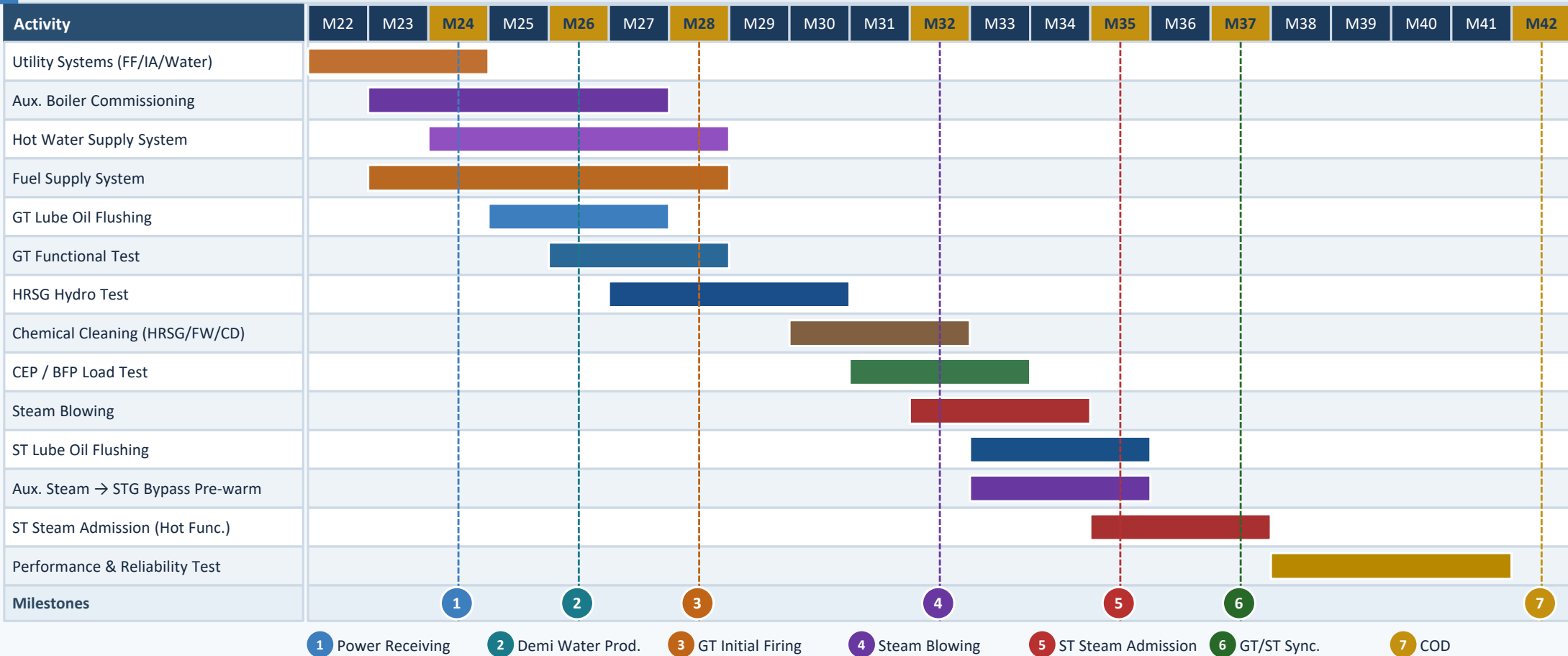
Overall EPC Project Timeline

Construction and commissioning schedule from NTP to Commercial Operation Date (COD)



Commissioning Phase Timeline

Detailed activity schedule showing Aux. Boiler, HWS, and all pre-commissioning piping milestones



Piping System Completion Priority by Commissioning Stage

Each commissioning milestone is gated by specific piping system readiness

◆ Power Receiving

- Fire Fighting — U/G & A/G complete, hydrants operable
- Potable & Service Water — WT/WWT & CEB restrooms
- Instrument Air — all distribution purged & pressure-tested
- Aux. Boiler piping — initial steam supply circuit ready

◆ GT Initial Firing

- Fuel Supply — cleaned, N₂ purged, all NDE complete
- GT Lube Oil — flushed; CCW lube oil cooler operational
- Aux. Boiler Steam → HWS heat exchanger (anti-icing loop)
- Hot Water Supply — to GT Air Intake anti-icing circuit
- GT Exhaust — expansion joints, insulation, supports done

◆ Steam Blowing

- All steam piping hydro-tested & insulated
- Aux. Boiler Steam — for STG Bypass System pre-warm
- Bypass piping operable; temp. silencers secured

◆ Demi Water Production

- Raw water intake → Water Treatment Building piping
- Demi. Water tank, pumps & distribution headers
- Hot Water Supply (HWS) piping — to HWS skid ready

◆ HRSG Hydro Test

- All HP/IP/LP steam piping installed and NDE-complete
- BFP installed and aligned; temp. manifolds ready
- Demi. water supply to HRSG confirmed operational

◆ ST Steam Admission

- Main steam (HP/LP) aligned, tested & insulated
- Sampling & chemical dosing lines in service
- Aux. Boiler Steam — standby for gland sealing backup

Utility Systems — Foundation of CCPP Construction

Installed first to support construction activities and all subsequent commissioning work

Raw Water Supply

- Supplied from external source (municipal / river intake)
- Primary resource for Demi. Water production and Fire Fighting supply
- HDPE or PE-coated steel U/G piping to WTP; SS transfer lines
- Must be completed on high-priority basis — pre-requisite for demi. water

Fire Fighting System

- Hydrants, hose reels, sprinklers, deluge systems
- Must be operational before power receiving (safety mandate)
- HDPE underground mains; galv. carbon steel aboveground

Demi. Water System

- High-purity water for HRSG, turbines & system flushing
- Critical for HRSG hydro test & chemical cleaning circuits
- S/S or HDPE piping from WTP → storage tanks → distribution

Potable Water System

- Drinking water and sanitation for plant personnel
- Eye shower & safety deluge stations at chemical handling areas
- Required at Occupied building before power receiving

Service Water System

- Non-potable water for equipment cooling & washing
- Supplied to chemical dosing areas, cooling tower & misc.
- S/S or PE-lined piping with isolation & flow control valves

Fuel Supply

- Fuel gas/oil from OSBL to GT skids
- 100% NDE on butt welds; pig/acid cleaning mandatory
- Vent lines to elevated safe discharge points

Instrument Air

- Clean, dry, oil-free air for pneumatic controls & actuators
- Completed and purged before power receiving
- Distribution along pipe racks to all skids & panels

N₂ Gas System

- Purging, blanketing & pressurization across plant systems
- Used for pipe preservation after chemical cleaning
- Regulators, isolation valves & check valves throughout

Systems Required Before GT Initial Firing

All systems below must be installed, tested, and commissioned before first GT ignition

Fuel Supply System

- Gas from governor station → performance heaters → GT manifold
- High-pressure welds: 100% NDE; acid/pig cleaning mandatory
- N₂ purge before fuel introduction; vent lines to safe discharge

Lube Oil & CCW System

- Lubricates turbine bearings, gearboxes & rotating equipment
- Return lines: proper slope for smooth drainage to reservoir
- CCW supplies lube oil cooler to maintain safe oil temperature

GT Air Intake & Anti-Icing

- Supplies clean, debris-free air to GT compressor
- Winter operation: Hot Water from HWS loop prevents icing on inlet guide vanes
- Compact low-pressure-drop layout; inlet elevation minimizes debris

Instrument Air & N₂

- IA: pneumatic valves, actuators, all control devices
- N₂: purge fuel lines, blanket storage tanks & sensitive areas
- Both: pressure relief, leak detection, monitoring instruments

GT Exhaust System

- Exhaust diffuser → insulated ducting → HRSG or bypass stack
- Expansion joints for thermal growth; spring hangers for movement
- High-temp insulation; diverter damper for simple/combined cycle switch

GT Misc. / Drain Systems

- Fuel oil/gas vent lines to elevated discharge (flame arrestors)
- GT/ST lube oil vent with mist eliminators
- Drain & wastewater transfer for safe effluent handling

Aux. Steam (AS) System

- Produces steam before HRSG becomes operational — independent of GT/ST cycle
- Supplies Aux. Steam as heat source to HWS heat exchanger → hot water for GT Air Intake anti-icing in winter
- Provides Aux. Steam to pre-warm STG Bypass piping & valves before Combined Cycle first steam admission

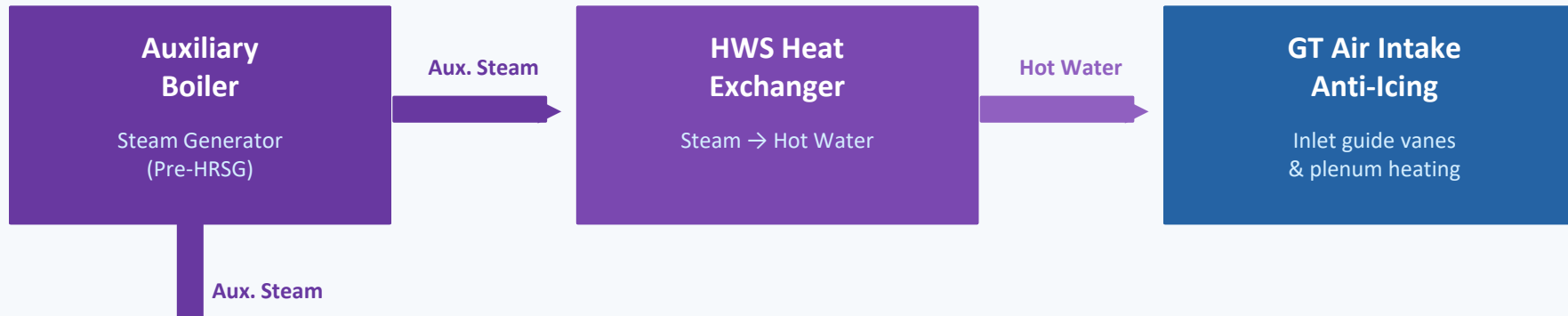
Hot Water Supply (HWS) System

- Distributes hot water to GT Air Intake anti-icing loop — prevents icing in winter operation
- Heat source (Pre-CC): Aux. Boiler Steam → shell & tube HX → hot water loop to inlet guide vanes & plenum surfaces
- After CC operation: HRSG extraction steam replaces Aux. Boiler as heat source; Aux. Boiler remains on standby

Aux. Boiler Steam & Hot Water Supply — System Concept

Heat flow: Aux. Boiler → HWS HX → GT Anti-icing | Aux. Steam → STG Bypass Pre-warm

PHASE 1 — Simple Cycle / Pre-firing : Aux. Boiler is primary heat source



PHASE 2 — Combined Cycle Operation : HRSG extraction replaces Aux. Boiler; Aux. Boiler on standby



Key Concept:

- ① Aux. Boiler generates steam before HRSG is operational — independent heat source for early commissioning.
- ② Aux. Steam → HWS Heat Exchanger → hot water prevents GT Air Intake icing in winter before simple cycle firing.
- ③ Aux. Steam pre-warms STG Bypass piping & valves before CC first steam admission to prevent thermal shock.
- ④ After CC commissioning: HRSG extraction steam takes over HWS heating; Aux. Boiler switches to standby mode.

Combined Cycle Phase — HRSG & Steam Turbine Piping Systems

Core systems linking GT exhaust to steam power generation

HRSG Steam System (HP / LP)

- Multi-pressure steam at high, intermediate, and low levels
- Expansion joints, spring hangers & constant-load hangers
- All welds verified RT/PAUT; hydro test before commissioning

Feed Water System

- Condensate → polishing → deaerator → HRSG inlet
- High-pressure delivery with thermal expansion provisions
- Hydro test coordinated with HRSG steam circuit testing

Condensate Water System

- Condenser hot well → LP heaters → deaerator
- Minimizes treated water loss; preserves water chemistry
- Typically routed below turbine floor on dedicated pipe racks

STG Bypass Steam System ★

- Diverts steam to condenser during start-up and shutdown
- BEFORE CC: Aux. Boiler Steam pre-warms bypass piping & valves — prevents thermal shock on first live steam admission
- After CC: HRSG steam takes over; bypass valves controlled by DCS

Circulation Water (CW) System – Not Required

- GRP underground + rubber-lined CS aboveground piping
- Robust supports due to large bore diameter and heavy weight
- Expansion joints with tie-rods installed downstream of pumps

Sampling & Chemical Dosing

- Continuous water/steam chemistry monitoring at key points
- PH control (ammonia/amine), oxygen scavengers, phosphate dosing
- S/S tubing routed to centralized sampling panel with coolers

Plant Operation & Maintenance Systems

Lube Oil Treatment System

- Lube oil purification & centrifuge for GT/ST bearing oil; oil/water separator & mist collection
- Not required for GT Initial Firing — must be completed before Combined Cycle operation & COD

Waste Water Treatment System

- WWT plant including effluent treatment, chemical neutralization & discharge control
- Not on critical path for GT/ST commissioning — must complete before COD (regulatory & environmental)

Pre-Commissioning Activities — Piping Readiness Checklist

Critical activities that must be completed to support each commissioning stage

HRSG Hydro Test

- All HP/IP/LP circuits completely filled with demi. water
- Temporary manifolds, high-point vents & low-point drains installed
- Check valves at stop valve locations temporarily removed
- Spring supports set to cold position; calibrated gauges fitted

Steam Blowing

- All steam piping hydro-tested, dried & fully insulated
- Bypass steam piping tested for tightness and valve operability
- Temporary silencers & exhaust stacks secured against vibration
- All temp. lines removed before live steam admission to turbine

Chemical Cleaning

- Temporary circulation loops connecting chemical tanks to HRSG
- BFP installation completed before cleaning commences
- Acid cleaning: steam/feedwater/fuel systems
- Alkali cleaning: condensate systems; DI water rinse to target chemistry

GT & ST Lube Oil Flushing

- Lube oil piping connected: reservoir → bearings → coolers → filters
- Temporary flushing lines & bypass installed for high-velocity flow
- CCW to lube oil cooler operational to maintain oil temperature
- Debris capture via strainers; flush acceptance per NAS cleanliness

Key Takeaways for Piping Construction Success

01

Piping is the Critical Path

- Piping starts only after civil, structural & equipment prerequisites are done
- Experienced piping managers are essential to prevent schedule delays.
- Cross-departmental coordination is a critical competency for this role.

02

Follow the Sequence

- Never skip or reorder the milestone sequence.
- Hydro test must be completed before commissioning begins.
- Aux. Steam pre-warm is mandatory before first live steam into STG Bypass.

03

ASME Code is Non-Negotiable

- WPS/PQR/WQT required for every weld — no exceptions.
- NDE results must be approved before moving to the next stage.
- PWHT requires calibrated equipment and full TC documentation.

04

Manage Each Stage Thoroughly

- Stages in order: pipe/support install → pressure test → flushing → chemical cleaning.
- Each stage must be formally signed off before the next begins.
- Any shortcut or oversight can directly impact commissioning safety and schedule.

05

Steam Piping Special Care

- Install HP steam piping starting from the turbine nozzle outward.
- Never apply cold pull at any connection — ever.
- All field modifications must have formal engineering approval.
- One mistake in steam piping can stop the entire project.

06

Document Everything

- NDE reports, PWHT charts, and test certificates must be filed for all welds.
- Material traceability records are mandatory for every component.
- Complete as-built documentation is required for COD handover acceptance.

Piping Construction Management

Structured approach to code compliance, quality, safety, and schedule control

Code Compliance

- ASME B31.1 Power Piping Code — mandatory for all power plant piping
- WPS/PQR/WQT approved and maintained for all welding activities
- Critical for P91, P22, P11: temperature, preheat, PWHT requirements

Drawing & Material Control

- Continuous coordination with engineering for latest revision status
- Dedicated segregated storage for tagged critical items (valves, etc.)
- BOM-based receiving inspection and heat number traceability

System-Based Priority

- Utility first → pipe rack headers → main process systems
- Large-bore and high-volume systems installed in bulk initially
- Commissioning sequence drives final installation priority order

Manpower Planning

- Secure experienced alloy/steam piping supervisors (foremen) early
- Dedicated scaffolding team runs concurrently with piping works
- Hydro test teams, equipment & common tools pre-mobilized

NDE & PWHT Compliance

- RT/PAUT on critical welds per ASME B31.1 acceptance criteria
- PWHT applied for alloy steels — calibrated equipment & TC records
- All NDE and PWHT results documented before proceeding to next stage

Schedule Risk Control

- Complete hydro tests before commissioning wherever possible
- Install isolation valves at building inlets for post-comm tests
- Vent & drain line requirements confirmed against test packages early

Underground Piping Works — Material & Installation Methods

Material-specific methods and quality requirements for buried pipe systems

HDPE

Fire water, potable water, raw water lines

- Butt fusion or electrofusion — fully homogenous, leak-proof joints
- Well-compacted granular bedding; anchor blocks at all bends & tees
- Hydrostatic pressure test before backfill; all fusion logs documented
- NFPA / AWWA or FM/UL-certified materials required

GRP (Not Required)

Circulating water, chemical-resistant lines

- Adhesive-bonded bell-and-spigot or laminated joints
- Uniform bedding; thrust blocks at all direction changes
- Risk: pipe sagging during backfill — restrain with chain blocks
- Coordinate U/G ↔ A/G alignment before concrete pours are placed

PE-Coated Steel

High-pressure water, fuel, process lines

- Welded joints; field joint coating via heat-shrink sleeves / epoxy
- Holiday detection test confirms full coating continuity
- Cathodic protection considered per site soil resistivity
- Air trap drain points at all low elevations — prevent condensate

CPVC

Chemical service water underground systems

- Solvent-cemented joints; ambient temperature curing essential
- Sand/fine granular bedding to prevent point loading & thermal stress
- Mechanical anchoring or encasement at direction changes
- Hydrostatic test before initial backfill layers are placed

⚠ When U/G piping connects to A/G piping without an isolation valve, A/G hydrostatic test must include the U/G section — plan isolation valves accordingly.

Aboveground & Steam Piping — Materials & Critical Inspection

High-temperature and high-pressure systems require specialized installation methods

Material Classes — Installation Requirements

Alloy Steel (P91 / P22 / P11)

Preheat + inter-pass temperature monitoring mandatory; PWHT required; RT/UT on all critical welds; stress analysis-compliant expansion loops

Carbon Steel

Dimensional check before erection; NDE on code-required joints; hydro test followed by surface treatment with primer and topcoat paint

Stainless Steel

Non-ferrous lifting gear; wooden sleepers for storage; full pickling & passivation of weld seams; chloride-free water for hydrostatic test

Galvanized / PE-Coated Steel

Padded slings during handling; field welds recoated immediately with zinc spray; threaded joints sealed with approved jointing compound

Steam Piping — Critical Inspection Checklist

- ✓ Material Verification — heat numbers, MTRs, pressure class confirmed
- ✓ Dimensional & Alignment — slope, spacing at turbine / HRSG nozzles
- ✓ Welding QC — WPS/PQR/WQT; preheat & inter-pass temperature control
- ✓ NDE Compliance — RT, UT, MT, PT per code; documented before proceeding
- ✓ PWHT Application — per ASME B31.1 / WPS; calibrated TC records
- ✓ Support & Hanger — spring hangers, snubbers: cold/hot settings verified
- ✓ Expansion & Flexibility — loops / offsets per stress analysis design
- ✓ Hydro Test Prep — temp vents/drains; calibrated gauges; code pressure
- ✓ Post-Test — drain, dry & nitrogen purge before steam blowing commences

Pressure Testing & PWHT Requirements

ASME B31.1 / B31.3 compliance for pressure testing and post-weld heat treatment

Hydrostatic Testing (Preferred)

Test Pressure	≥ 1.5× design pressure (ASME B31.1 & B31.3)
Hold Time	Minimum 10 min. at test pressure (B31.1 requirement)
Test Medium	Clean non-corrosive water; temperature ≥ 10°C
Pre-test	NDE & PWHT complete; non-rated items isolated; springs at cold set
Post-test	Gradual depressurization; complete drain & dry; N ₂ purge if needed

Pneumatic Testing (Special Cases Only)

Test Pressure	1.1 to 1.33× design pressure; stress ≤ 90% yield
When Permitted	Only if liquid test impractical (Category D service)
Risk Assessment	Formal RA + written method statement mandatory
Safety	Relief devices set below max; non-essential personnel excluded
Monitoring	Pressure-time chart; barriers; gradual incremental steps

Post-Weld Heat Treatment (PWHT) — Requirements per ASME B31.1

When Required

- P-No.1 Carbon Steel: wall ≥ 19mm (¾")
- P-No.3 (P11, 1.25Cr): ≥ 19mm; mandatory if C > 0.15%
- P-No.4 (P22, 2.25Cr): ≥ 13mm
- P-No.5 (P91, 9Cr): ALWAYS — regardless of thickness

Temperature Ranges

- P-No.1 Carbon Steel: 595 – 675 °C
- P-No.3 (P11): 595 – 720 °C
- P-No.4 (P22): 675 – 760 °C
- P-No.5 (P91): 730 – 800 °C (critical — closely monitored)

Execution Method

- Electric resistance heating pad or induction coil
- Min. 1 calibrated thermocouple per 91 cm weld length
- Heating / cooling rates controlled per WPS
- Temperature recorder with continuous chart record

Documentation Required

- PWHT time-temperature chart — signed by QC Inspector
- Hardness test after PWHT (P91: max 265 HB per ASME)
- PWHT report referenced in weld map and NDE dossier
- All records included in turnover / handover package

Pipe Support Installation — Types & Critical Rules

Properly designed and installed supports prevent over-stress, vibration, and thermal damage

Rigid Supports

Used at anchors, guides, and rest points. Transfer static and dynamic loads directly to structure. No flexibility — must be positioned exactly per stress drawings.

Spring Supports

Variable spring hangers accommodate vertical thermal displacement while maintaining load transfer. Set to cold load during erection; both cold & hot settings documented.

Snubbers

Protect piping from sudden dynamic loads: seismic, water hammer, turbine trips. Allow slow thermal movement but restrain rapid displacement. Must remain accessible post-insulation.

Sliding Supports

Controlled axial movement while restricting lateral displacement. Steel plate or PTFE pad with precise alignment and lubrication. Verify friction vs. thermal force balance.

Critical Rules — Steam Piping Support Installation

- Install HP steam piping starting at the turbine nozzle and work outward — NEVER apply cold pull under any circumstance
- Follow approved stress analysis drawings exactly — field modifications require formal engineering approval before execution
- Spring supports: adjust to cold setting during erection; mark and document both cold and hot position values
- After hydrostatic test: perform cold-setting adjustment before steam blowing (cold-set may shift during pressurization)
- During steam admission (Hot Function): TA confirms that piping thermal movement direction matches stress analysis predictions
- All spring adjustment points and inspection mechanisms must remain fully accessible after insulation cladding is installed

Lessons & Learned — HP Turbine Rubbing Incident

Real case study: HP turbine damage caused by improper steam piping support design and installation

INCIDENT: Sudden high vibration during Performance Test → Heavy rubbing inside HP Turbine → Rotor contact with diaphragm → Turbine disassembly required

Root Cause Analysis

- Official: HP Steam piping thermal expansion constrained at MSV connection — improper support design and missing stress analysis factors
- Potential: Incorrect thermal expansion calculation or violation of pre-warming (warm-up) procedures
- Potential: Cold pull applied during HP steam pipe installation at turbine nozzle connection
- Potential: Misalignment of shaft, bearings, or components due to poor installation

Corrective Actions

- Full disassembly, inspection, repair and reinstallation of HP Turbine by OEM
- Redesign and modification of HP Steam piping supports to fully accommodate designed thermal movement
- Improvement of turbine pre-warming and commissioning operational logic
- Re-verification of all support cold-settings and stress analysis compliance as-built

Preventive Measures

- ALWAYS install HP Steam piping from the turbine nozzle outward — never apply cold pull to close a gap
- Strictly comply with all support drawings: gap clearances, cold-setting values, and design load ratings
- Confirm that stress analysis assumptions match the actual as-built piping configuration
- TA (Turbine OEM Technical Advisor) must witness cold-setting and thermal movement checks during steam admission
- Monitor and record piping movement directions during Hot Function — report any deviation from design immediately